

PAPER 1 / QUESTION 115

2225-6707 / original question 4

115. A sphere of density ρ and radius R rests on the bottom of a tank of water. The buoyancy force on the sphere is F_1 .

A sphere of density 2ρ and radius $\frac{R}{2}$ is at the bottom of the same tank. The buoyancy force on the second sphere is F_2 .

What is $\frac{F_1}{F_2}$?

- A. 2
- B. 4
- C. 8
- D. 16